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Evaluating the necessity of the multiple metrics for assessing explainable AI: A critical examination

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ABSTRACT

This paper investigates the specific properties of Explainable Artificial Intelligence (xAI), particularly when implemented in AI/ML models across high-stakes sectors, in this case cybersecurity. The authors execute a comprehensive systematic review of xAI properties, various evaluation metrics, and existing frameworks to assess their utility and relevance. Subsequently, the experimental sections evaluate selected xAI techniques against these metrics, delivering key insights into their practical utility and effectiveness. The findings highlight that the proliferation of metrics enhances the understanding of xAI systems but simultaneously exposes challenges such as metric duplication, inefficacy, and confusion. These issues underscore the pressing need for standardized evaluation frameworks to streamline their application and strengthen their effectiveness, thereby improving the overall utility of xAI in critical domains.

1. Introduction

1.1. Background and motivation

The technologies of Artificial Intelligence (AI) and Machine Learning (ML) have become ubiquitous across a wide range of industries, revolutionizing practices in diverse sectors ranging from healthcare, finance and autonomous driving to disinformation detection and cybersecurity [1,2]. In the context of cybersecurity, the integration of AI/ML has introduced advanced tools and methodologies for detecting, analysing, and responding to threats with unprecedented speed and efficiency [3,4]. This integration underscores the growing reliance on these technologies to safeguard digital infrastructures and sensitive data against increasingly sophisticated cyber threats [5].

However, besides bringing significant advancements, the widespread adoption of AI/ML models emphasizes a critical challenge: the opacity of these models [6]. This lack of transparency may be problematic in every domain; nevertheless, in high-stakes domains, the decisions made by AI systems can have particularly far-reaching consequences [7]. The "black box" nature of many AI/ML systems, where the decision-making processes are not readily understandable by humans, poses significant risks [8,9]. It abuses trust and accountability, making it difficult to diagnose errors, understand predictions, or justify decisions to stakeholders [10].

In order to mitigate the opacity inherent in many AI/ML models, the concept of Explainable AI (xAI) emerges as a crucial innovation, providing a way to achieving transparency, understandability, and interpretability in complex decision-making processes [11]. At the heart of xAI there are several pioneering techniques designed to uncover the layers of sophisticated AI models, allowing for insight into their operational mechanics. Among these, Anchors offer a model-agnostic approach by pinpointing rules or "anchors" that provide precise explanations for the predictions on specific instances, thus offering clear, rule-based insights into model behaviour [12]. Local Interpretable Model-agnostic Explanations (LIME) dissect model predictions by constructing surrogate models that are interpretable over the local prediction vicinity, thereby shedding light on how the model behaves around the area of interest [13,14]. SHapley Additive exPlanations (SHAP) leverages the concept of Shapley values from cooperative game theory to fairly distribute the "payout" (prediction) among the features, treating each feature value as a "player" in the game. This method offers a unified measure of feature importance that combines consistency with local accuracy [13,15]. Integrated Gradients

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(IG), on the other hand, attributes the prediction of a neural network to its input features in a way that is both detailed and mathematically grounded, using gradients to trace the contribution of each input to the final decision [16]. Each of these techniques approaches the challenge of explainability from a unique angle, whether by highlighting the conditions under which a model makes certain decisions, approximating local decision boundaries, or distributing the responsibility of a prediction across input features.

Despite the advancements in xAI and the development of techniques to enhance model transparency, the field still faces the challenge of effectively evaluating explainability. The absence of a unified standard or consensus among researchers on how best to measure and evaluate the quality of explanations remains a significant problem. The scholarly community continues to propose new metrics, each aiming to capture different aspects of explainability. This ongoing proliferation of metrics reflects the complexity of the task at hand, highlighting the need for a systematic approach to evaluate and consolidate these efforts into a coherent framework that can universally assess the effectiveness of xAI applications.

Consequently, the aim of the study presented has been to examine the properties of explainable AI by systematically identifying them, cataloguing specific measures of explanations as proposed by researchers, and outlining the existing frameworks and taxonomies. Subsequently, the study advances by applying these metrics in an experimental context to evaluate the effectiveness of various xAI methods, thereby bridging the gap between theoretical evaluation and practical applicability.

1.2. Contributions

This paper makes several contributions to the field of xAI research. Firstly, it offers a comprehensive synthesis of the properties of explainability, gathering and analysing them, as well as collecting diverse measures and metrics proposed within the academic community, along with existing taxonomies and frameworks for evaluating explanations. This consolidation provides clarity and direction for future research and practical application in xAI. It also highlights that the domain of xAI metrics is vast, marked by a diversity of metrics and an apparent lack of consensus on fundamental properties. Secondly, through its systematic approach and experimental validation, the study contributes to the standardization of evaluation methods in xAI, opening the doors to more consistent and reliable measures of explainability across different domains and applications. The unique value of including empirical results alongside the theoretical ones is that they help identify context-specific performance, ensuring that practitioners choose the most appropriate metrics for their specific needs.

1.3. The structure of this paper

This work is structured as follows. Starting with an introduction that lays the groundwork, the authors present the study's background, motivation, and key contributions, setting the stage for the comprehensive study that follows. Section 2 describes the systematic review approach, specifying details such as the research framework, selection criteria, and data extraction method. The following Section 3 then systematically examines the properties and metrics of explainability in AI, leading into a review of evaluation frameworks and taxonomies in the "xAI Evaluation Taxonomies and Frameworks" section. Then, Section 4 describes the practical applications of the study, including methodology and metrics, with the results of the experimental part discussed in Section 5. The paper culminates in the final Section 6, presenting the findings and discussion before concluding with a summary of insights and implications for future research.

2. Survey structure and methodology

This section discusses the structured approach and methodology implemented for the survey part of the paper, providing a bird's-eye overview of the foundational framework of the investigation. It presents the formulation of research questions, identification of relevant databases, the development of search strings, inclusion and exclusion criteria, as well as the process of data extraction, collectively forming the structure of the exploration into the evaluation of explainable AI metrics.

2.1. The course of the study

The course of the study presented was as follows. Firstly, the study's research questions were established. Then, a search string was developed to identify relevant studies. This was followed by the execution of the literature search. Afterwards, criteria for including and excluding studies were determined and subsequently applied to the identified papers. The papers were then systematically classified. Following this, relevant data was collected from these papers. Finally, the compiled data was examined to respond to the research questions.

2.2. Research questions

In addressing the complexities of xAI, especially in the context of cybersecurity, this section introduces three areas of investigation: identifying the essential properties of xAI explanations as highlighted in literature, examining the quantitative metrics proposed for evaluating explanation quality, and exploring existing frameworks that provide structured approaches for xAI assessment. Thus, the ultimate Research Questions posed for the study presented have been as follows:

RQ1. *What properties of explanations are identified as significant in the literature for ensuring the effectiveness of xAI systems?*

This question aims to compile and categorize the various characteristics of explanations essential for xAI, as outlined in academic and scientific research.

RQ2. *Which metrics have been proposed to quantitatively assess the quality of explanations in xAI?*

Seeking to gather and evaluate the different quantitative measures developed or applied by researchers to assess explanation quality, this question mostly includes metrics believed to be objective, as categorized by [17].

RQ3. *What taxonomies or frameworks exist for evaluating xAI systems, and how do they categorize and measure explanation quality?*

Finally, this question explores the existence of structured approaches or models for the assessment of xAI, focusing on how these frameworks classify explanation properties and the methods they recommend for evaluation.

The answers to these questions have been presented in Sections 3, 3.14-3.15 and 3.16, respectively.

2.3. Bibliographic methodology

This study employed a systematic literature review method, guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework, as outlined by Page et al. (2021) [18]. The methodology was designed to ensure a thorough analysis of high-quality research papers selected from prestigious databases.

Consequently, the databases used in this study, i.e., Science Direct, IEEE Xplore, DBLP, and arXiv, were selected as they are well-regarded for their comprehensive collections of peer-reviewed articles in the fields of computer science and technology, especially regarding artificial intelligence and cybersecurity. These sources are acknowledged for hosting significant and leading-edge research, crucial for gathering the most relevant and influential studies for this review. Further exploration through ResearchGate allowed for the acquisition of insights

and recent discoveries directly from researchers and professionals, adding depth to the review. A supplementary search using Google was executed to include possible additional documents, ensuring a broad and varied range of materials for analysis.

This detailed approach to sourcing literature aimed to establish a solid basis for the research outcomes.

2.4. Search string development

In order to develop the search string for the literature study, the PICO methodology [19] was applied.

The PICO (Population, Intervention, Comparison, Outcome) methodology is a structured framework that guides the development of research questions and search strategies in evidence-based practice. For the sake of this study, the elements of the search strategy were developed according to the framework, in the following manner:

- Population: AI/ML models or systems.
- Intervention: Implementation of xAI techniques or tools.
- Comparison: In this case, this was not directly applicable, as the focus is on identifying metrics rather than comparing interventions. However, it did not affect the study, because, as remarked by [20], in systematic review studies, the Comparison and Outcome criteria are optional.
- Outcome: Metrics, evaluation methods, frameworks, or taxonomies for assessing the explanations provided by xAI.

Given the above considerations, a comprehensive search string designed to find literature on xAI metrics, evaluations, frameworks, and taxonomies was developed taking into account:

1. Keywords related to AI/ML models or systems: (“artificial intelligence” OR “AI” OR “machine learning” OR “ML”)
2. Intervention with xAI: (“explainable AI” OR “xAI” OR “interpretable AI”)
3. Outcomes in the form of metrics and evaluation: (“metrics” OR “evaluation” OR “assessment” OR “measurement” OR “frameworks” OR “taxonomies” OR “methodologies” OR “approaches”).

Combining these elements, the ultimate search string was:

(“artificial intelligence” OR “AI” OR “machine learning” OR “ML”) AND (“explainable AI” OR “xAI” OR “interpretable AI”) AND (“metrics” OR “evaluation” OR “assessment” OR “measurement” OR “frameworks” OR “taxonomies” OR “methodologies” OR “approaches”).

This search string was designed to be broad enough to capture a wide range of relevant studies while being specific to the aspects of xAI the authors were interested in exploring.

2.5. Inclusion and exclusion criteria

To ensure the selection of the most relevant and valuable research articles, the authors established specific criteria for inclusion and exclusion. These criteria were developed through the combined expertise of the study’s authors and a comprehensive review of several high-quality survey papers in similar domains.

The established criteria are detailed below.

Inclusion Criteria:

- Papers published in the last 10 years: To ensure the research is current and reflects recent developments in the field.
- Papers from journals, conferences, or workshops: These are typically peer-reviewed sources, indicating a level of quality, credibility and reliability.
- In English: To ensure accessibility and comprehensibility for the research team and the broader academic community.

Exclusion Criteria:

- Papers published before 2014: To maintain the currency of the research data and perspectives.
- Books, book chapters, and non-scientific papers: These are excluded due to varying levels of scrutiny and peer review compared to journal articles and conference papers.
- Studies not available in full-text: Full access is required for a thorough review and analysis.
- Non-English papers: Language constraints limit the review to English-language documents only.
- Duplicates: To eliminate redundancy and ensure each selected study contributes unique insights.
- Non-peer-reviewed material: Ensuring that all reviewed material has undergone a rigorous, standardized quality check.

These criteria were applied initially to abstracts and metadata, and where necessary, to the full text of the papers.

On top of that, the selection process incorporated a refinement stage where the authors could, following a group discussion, decide to include a paper that was found to be exceptionally relevant despite not meeting all the established inclusion criteria (e.g., if the authors found an older paper of exceptional quality and/or relevance for the study, they would include it in it). This approach allowed for flexibility in the paper selection process, ensuring that the research contributing significantly to the study’s objectives was not overlooked.

The initial search of the databases yielded 6137 potentially relevant results altogether. Having applied the inclusion and exclusion criteria, 94 papers were selected. Then, after this initial round of screening was completed, the researchers were left with 47 papers dealing with xAI evaluation metrics, which were then subject to further analyses.

2.6. Data extraction

Following the selection of the scientific papers for in-depth data analysis, the research team determined the specific information to be extracted and recorded on the data extraction form. It was mutually decided that the one of the authors would undertake the task of extracting the data, with the rest of the team responsible for reviewing the accuracy and completeness of this process.

To adequately address Research Questions 1-3, it was essential to identify distinct properties of explanations, any specific metrics (or suggestions of such metrics), and any frameworks or taxonomies related to the evaluation of explanations. Additional information extracted included the authors’ names, publication year, paper title, and any other relevant pieces of data.

The findings from this data extraction effort, including responses to the Research Questions and overall conclusions, will be detailed in the subsequent sections of this paper.

3. The results of the systematic review study: the properties of explainability, and evaluation metrics and frameworks

The following section presents the outcomes of the systematic review study, addressing Research Questions 1-3. Through analysis and evaluation, the authors have distilled the findings from the selected literature to offer comprehensive insights into the properties of explanations, metrics, and frameworks or taxonomies for evaluating explainable AI systems. These results aim to enrich the current understanding and contribute significantly to the field, providing clear answers to the initially posed research questions.

The properties of explainability

This section provides answers to the Research Question 1, “What properties of explanations are identified as significant in the literature for ensuring the effectiveness of xAI systems?”. As it turned out in the course of the study, researchers have proposed a wide variety of properties they believe are necessary for measuring explainability. This section discusses these properties as described in the target literature, from the most to the least mentioned ones.

3.1. Fidelity

The property which gets mentioned most often is fidelity. Zhou et al. highlight that fidelity is composed of the aspects of completeness and soundness. In this context, completeness means that the explanation covers all parts of how the model works. Soundness relates to how accurate and true the explanation is [21]. Guidotti et al. see fidelity as a measure used to evaluate how closely an interpretable predictor, *c*, can replicate the behaviour of the original, more complex predictor, *b*, on a test dataset. In practical terms, fidelity is measured by how often $c(x)=b(x)$ for each data point in the test dataset, effectively showing how accurately predictor *c* can mimic the predictions of the original model *b* [22]. Molnar et al. describe fidelity as the accuracy with which an explanation reflects the ML model [17]. Lakkaraju et al. suggest that fidelity should accurately represent the behaviour of the complex, or "black box", model across every feature of the dataset. This can be achieved when the predictions made by the simpler model match those of the complex model for as many data points as possible. To measure this, they introduce a way to calculate fidelity by looking at how often the simpler model's predictions disagree with the complex model's predictions [23]. Markus et al. note that explanations based on models automatically meet the requirement of completeness because they provide enough information to predict the output for any given input. Soundness is assured when the model used for explanation is the same as the task model, as seen in explainable modelling. According to them, for explanations that come after the model has been created, the fidelity measure, as defined by Lakkaraju et al. helps determine how well the explanation matches the original model's predictions by looking at the percentage of predictions that are the same [24].

3.2. Faithfulness

Faithfulness, as described by Alvarez-Melis, inquires whether the relevance scores assigned by explanations accurately reflect the genuine importance of features in a model's decision-making process [25]. This concept is alternatively termed fidelity within the context of the OpenXAI project. To evaluate the precision with which an explanation replicates the underlying model, previous studies have employed synthetic datasets to derive clear, accurate explanations or have analysed how variations in feature values impact predictions. OpenXAI distinguishes two primary types of metrics for assessing faithfulness: ground-truth and predictive faithfulness. Ground-truth faithfulness relies on definitive benchmarks to appraise explanations, whereas predictive faithfulness examines the extent to which the explanation identifies features that significantly influence the model's predictions. This implies that features identified as more important should have a greater influence on the outcomes of the model [26].

3.3. Interpretability

When it comes to interpretability, Zhou et al. propose that interpretability can be characterized by three essential properties: clarity, broadness, and parsimony. Interpretability's clarity means that the provided explanation must be clear and free from ambiguity. Parsimony indicates that the explanation should be succinct, utilizing the simplest possible form without sacrificing necessary detail. Furthermore, an explanation's broadness, which assesses its general applicability, is also considered a crucial aspect of interpretability, suggesting that the best explanations are not only straightforward but also widely applicable [21]. In turn, Guidotti et al. define interpretability as the degree to which a model or its predictions can be understood by humans. A central part of this discussion focuses on determining measurable criteria for interpretability, with one proposed method being to evaluate the complexity of the predictive model, particularly in terms of its size. This concept of interpretability is sometimes also referred to as comprehensibility in the literature [22]. Lakkaraju et al. offer a quantitative approach to assess interpretability, emphasizing the simplicity and non-redundancy of decision sets. The first metric, *f1(R)*, rewards decision sets with fewer rules, using the size difference between the total input itemsets and the decision set. The second metric, *f2(R)*, prefers decision sets with rules that contain fewer predicates, calculating this based on the maximum possible rule length and the lengths of the actual rules in the set. Lakkaraju et al. also introduce criteria to reduce overlap in feature space among rules, aiming to make it easier for users to understand how attribute values correlate with class labels. This is quantified through two metrics, *f3(R)* and *f4(R)*, which penalize overlaps among rules, distinguishing between those that predict the same class and those that predict different classes. Lastly, a term *f5(R)* is included to ensure that the decision set encompasses at least one rule for each class, addressing the importance of representing all classes, especially rare ones, thereby balancing comprehensiveness with the necessity for brevity. This comprehensive framework underscores the importance of clear, applicable, and concise explanations in making complex models interpretable to users [23].

3.4. Completeness

Markus et al. summarize completeness by stating that an explanation should delineate the entire dynamic of the task model. This means providing enough information to predict the output for any given input, thereby achieving completeness. This comprehensive view of completeness underlines the importance of creating explanations that are detailed and accurate, yet digestible and pertinent, to effectively communicate the intricacies of complex models [21]. In turn, completeness, as seen by Zhou et al. is an integral aspect of fidelity. It ensures that an explanation comprehensively encompasses the dynamics of a machine learning model [21].

Nauta et al. expand on this concept by asserting that completeness gauges how well an explanation encapsulates the predictive model *f*. The ideal scenario is where the explanation conveys the "whole truth", providing detailed insights while maintaining a balance with brevity and accuracy to avoid overwhelming the audience. The concept of reasoning-completeness is introduced to describe the extent to which an explanation details the internal workings of the model. This can range from exposing every mathematical operation and parameter, characteristic of white-box models known for their inherent transparency, to employing global surrogate models that replicate the predictions of a black box model *f* without exploring its internal logic. The choice between levels of reasoning-completeness depends on the desired application and context, making it a qualitative measure for comparing different types of explanations. Output-completeness, on the other hand, focuses on how thoroughly an explanation covers the outputs produced by model *f*. It acts as a measure of how much of the model's predictive outcomes can be explained, essentially quantifying the features of the model's output that remain inexplicable. This dimension of completeness evaluates the congruence between the explanation method and the original model's predictions, aiming for a high degree of alignment.

3.5. Clarity

Markus et al. emphasize that clarity is also a critical aspect of interpretability, asserting that an explanation may be called interpretable when it is straightforward, meaning it presents a singular, consistent rationale applicable across similar instances [24]. Similarly, Zhou et al. highlight that clarity is essential for an explanation to be considered effective, underscoring the importance of it being free from ambiguity to ensure that the explanation's rationale is clear and easily understood [21].

3.6. Complexity

Guidotti et al. underscore that the complexity of a model, although critical, is often given limited attention in the literature. The evaluation of a model's complexity is closely linked to its comprehensibility, presenting a significant challenge due to the difficulty of accurately assessing complexity. Typically, this evaluation is approximated based on the model's size. Furthermore, complexity is frequently viewed as being inversely related to interpretability. When discussing the challenge of explaining "black box" models, it is noted that complexity pertains solely to the model itself, rather than the training data, which is usually not considered. Thus, different methods for quantifying complexity are mentioned. For example, one approach identifies complexity through the number of distinct regions within the model, each defined by its boundaries. For linear models, complexity may be gauged by the count of non-zero weights, and for decision trees, by the depth of the tree. Complexity can also be measured by the length of a rule's condition, defined by the number of attribute–value pairs it contains. When comparing two rules of similar frequency and accuracy, the rule with fewer condition elements is often preferred for its greater interpretability. In the context of rule lists, complexity is generally assessed by the total count of attribute–value pairs across all rules. However, this method may not always effectively measure model complexity, particularly for ordered rule lists, where different test records may require different numbers of rules for evaluation. An alternative, more accurate measure for such models might be the average number of conditions assessed to classify a set of test records, though this is considered more a measure of explanation than of the rule list's inherent complexity. Lastly, the authors of the Quantus framework introduce a perspective focusing on the conciseness of explanations, indicating that complexity is inversely related to how concisely a model's prediction can be explained, ideally using a minimal number of features. This approach aligns with the broader understanding that less complex explanations, requiring fewer features to explain a prediction, are preferable for enhancing interpretability [27].

3.7. Fairness

The authors of the OpenXAI framework emphasize the importance of fairness in the context of machine learning explanations, highlighting the need to eliminate significant disparities in the reliability of post hoc explanations across different demographic subgroups. This approach involves averaging various metric values for instances within both majority and minority subgroups and then comparing these averages to identify any notable disparities. The concept of fairness, as described by Agarwal et al. aims to ensure that explanations are equally faithful and stable, without any bias towards or against specific groups [26]. In contrast, Guidotti et al. touch on fairness from a different angle, focusing on the model's responsibility to prevent both direct and indirect discrimination against groups. This principle of fairness calls for measures within the model's design and application that safeguard against biases that could harm or disadvantage any group based on demographic or other distinguishing characteristics [22].

3.8. Monotonicity

Monotonicity has been mentioned as a significant property in the AI360 framework. Guidotti et al. define the principle of monotonicity in the context of predictors. A predictor adheres to this principle if, when the values of a numerical attribute increase, the probability of a record being classified as a member of a certain class either consistently increases or decreases in a monotonic fashion. This means that the relationship between an attribute's value and the outcome is either always positive or always negative, without any reversals in direction [22].

3.9. Parsimony

Parsimony, as outlined by Zhou et al. is considered a key aspect of interpretability. Parsimony indicates that an explanation should be conveyed in a straightforward and concise manner, emphasizing the value of simplicity. This approach ensures that explanations are accessible and easily understood, without unnecessary complexity. It is worth mentioning that Zhou et al. also refer to parsimony as "simplicity" [21]. Similarly, Markus et al. echo the sentiment that explanations should avoid being overly complex. They highlight the necessity of presenting explanations in a compact format, reinforcing the principle that the clarity and effectiveness of an explanation are enhanced when it is distilled to its most essential elements. This focus on simplicity aids in making the underlying mechanics of a model more transparent and comprehensible to a broader audience [24].

3.10. Robustness

Guidotti et al. articulate that robustness in a model signifies its capacity to sustain performance levels, regardless of minor fluctuations in its parameters or input data. This characteristic, synonymous with reliability, ensures the model's outcomes remain consistent even when faced with small changes in its operational environment [22]. The Quantus framework further refines this concept by focusing on the stability of explanations under slight input modifications, with the assumption that the model's output remains largely unchanged. This perspective on robustness emphasizes the importance of explanation consistency, particularly in how small variations in input do not lead to significant alterations in the explanations provided, thereby underlining the resilience of the model and its interpretative mechanisms [27].

3.11. Soundness

Zhou et al. describe soundness as a property integral to fidelity, emphasizing that it pertains to the accuracy and truthfulness of an explanation. This aspect focuses on how well the explanation aligns with the actual workings and outcomes of the machine learning model, ensuring that the insights provided are both correct and reflective of the model's true behaviour [21]. Markus et al. concur with this perspective, stating that for an explanation to be considered correct, it must be faithful to the task model. This means the explanation should accurately represent the model's decision-making process, providing a truthful and transparent account of how inputs are transformed into outputs [24].

3.12. Stability

Alvarez-Melis and Jaakkola emphasize the importance of stability in interpretability methods, defining it as the consistency of explanations for similar inputs. This consistency is crucial for generating meaningful explanations, especially when inputs undergo local perturbations. An illustration provided in their work demonstrates that popular interpretability methods fail to maintain stability, showing significant changes in explanations even with minimal additions of white noise to the input. To quantify stability, Alvarez-Melis and Jaakkola propose using a method to estimate the worst-case scenario of robustness to these perturbations. This involves calculating the maximum difference between explanations for an original input x and those for inputs within a small neighbourhood around x, effectively estimating the Lipschitz constant of the explanation generation model. This calculation can be straightforward for models like self-explaining neural networks (SENN), where explanations are directly tied to differentiable concepts, allowing for optimization through automatic differentiation and back-propagation. However, for post-hoc explanation frameworks, which lack end-to-end differentiability, this process presents significant challenges [25]. The authors of the OpenXAI framework build on this by adopting metrics introduced by Alvarez-Melis and Jaakkola, to

Fig. 1. The identified properties of xAI.

assess the robustness of explanations against minor input variations. They utilize metrics such as Relative Input Stability (RIS), Relative Representation Stability (RRS), and Relative Output Stability (ROS). These metrics are designed to capture the maximum change in the explanation relative to changes in the inputs, model parameters, and output prediction probabilities, respectively. Through these measures, OpenXAI aims to provide a comprehensive assessment of explanation stability, highlighting the resilience of explanations to variations across different aspects of the model's operation [26].

3.13. Other proposed properties of explainability

In addition to the properties defined above, various scholars have highlighted a wide spectrum of additional properties essential for the measurement of xAI's effectiveness. For the sake of brevity, this paper merely enumerates these properties, directing readers to the relevant literature for a deeper understanding.

Notably, Zhou et al. underscore the importance of broadness and explicitness [21], while Alvarez-Melis and Jaakkola emphasize intelligibility [25]. Guidotti et al. discuss accuracy, causality, generality, privacy, reliability scalability, usability [22], whilst Molnar et al. mention interaction strength, sensitivity to perturbations, simulatability, and sparsity [17]. Lakkaraju et al. bring interactivity and unambiguity into focus [23]. Nauta et al. introduce their own comprehensive list including coherence, compactness, completeness, composition, confidence, consistency, context, continuity, contrastivity, controllability, correctness, and covariate complexity [28]. Lastly, the Quantus framework presents localization and randomization as crucial metrics [27].

All the identified xAI properties, as proposed in the subject literature, have been illustrated in Fig. 1, with the size of the words reflecting how frequently a given xAI property has been mentioned in distinct studied sources.

The results of this analysis underscore the challenge within the field: despite the diverse nomenclature for properties of explainability, there is a notable lack of consistency in definitions, some level of overlap in distinct concepts, and a tendency for some authors to introduce unique properties that remain confined to their individual contributions. Addressing this issue is necessary for advancing the field.

3.14. xAI evaluation metrics

This section addresses the Research Question 2, "Which metrics have been proposed to quantitatively assess the quality of explanations in xAI?"

Consequently, in this section, the outcomes of the literature review have been presented, gathered in Table 1, which catalogues nearly ninety distinct metrics extracted from scholarly texts. It must be acknowledged that this compilation is not all-encompassing but rather gathers metrics recognized within academic discourse (e.g., survey papers), incorporated into existing xAI evaluation frameworks, or believed by the authors to be especially relevant to the objectives of this study.

The list of proposed metrics is not exhaustive; on the contrary, it includes only a selection of examples. The subject literature encompasses many more, with both new and repurposed metrics continually emerging. Even a short glance at the existing landscape reveals an overwhelming abundance of xAI metrics, underscoring a profound lack of agreement among researchers and practitioners regarding the fundamental properties these metrics aim to measure, similarly to the issue with xAI properties, mentioned earlier in this paper. This divergence not only highlights the multidimensional nature of explainability but also underscores the challenges in establishing standardized benchmarks for assessment.

These examples emphasize the significant challenge of the lack of uniformity in metric definitions in the field of xAI, which leads to ambiguity and potentially hinders the development of universally accepted standards for evaluating explanations. This fragmentation not only complicates the task of researchers and practitioners in selecting appropriate metrics for their needs but also makes it difficult to compare the results across different studies, ultimately stalling progress towards achieving interpretable and trustworthy AI systems. As evident form the presented material, the necessity for a concerted effort towards standardization, possibly through the establishment of a unified framework that accommodates the multifaceted nature of explainability, is both evident and urgent.

In addition to this, some researchers arrive at similar metrics independently, further complicating the domain. For example, the metric called completeness [64] has been said [27] to be identical or similar to at least three more metrics, Summation to Delta [71], sensitivity-n [65] and Conservation [38]. This emphasizes the need for a unified taxonomy.

3.15. Selected xAI evaluation metrics

This subsection introduces a selection of noteworthy xAI evaluation metrics, discussing the rationale behind their design, and illustrating the diverse approaches used to quantify explainability in AI systems.

Table 1
The compilation of the identified xAI evaluation metrics.

Name of the metric	Author(s)	Year	Source	What does it measure?
Attribution Localization	Kohlbrenner et al.	2020	[29]	localization [27], fidelity-soundness [30]
AUC	Fawcett et al.	2006	[31]	localization [27]
Average justification score	Laugel et al.	2019	[32]	faithfulness [32] fidelity-soundness [30]
Average rule length	Lakkaraju et al.	2016	[33]	interpretability [33] interpretability-simplicity [30]
Avg-Sensitivity	Yeh et al.	2019	[34]	sensitivity, (in)fidelity [34] robustness [27]
Completeness	Sundararajan et al.	2017	[16]	completeness [16,27]
Complexity	Ribeiro et al.	2016	[35]	complexity [35] interpretability-simplicity [30]
Complexity	Bhatt et al.	2020	[36]	complexity [27,36] interpretability-simplicity [30]
Consistency	Dasgupta et al.	2022	[37]	faithfulness [37] robustness [27]
Continuity	Montavon et al.	2018	[38]	robustness [27,38] interpretability-clarity [21,30]
Contrastive Explanations Method	Luss et al.	2019	[39]	monotonicity [40]
Correctness	Ignatiev	2020	[41]	correctness [41] fidelity-completeness [30]
Cover	Lakkaraju et al.	2017	[23]	unambiguity [23] interpretability-clarity [30]
Disagreement	Lakkaraju et al.	2017	[23]	fidelity [23] fidelity-soundness [30]
Diversity	Nguyen et al.	2020	[42]	simplicity [42] interpretability-simplicity [21,30]
Effective complexity	Nguyen et al.	2020	[42]	broadness, simplicity [42] complexity [27] interpretability-broadness and simplicity [30]
Efficient MPRT	Hedström et al.	2023	[43]	complexity, faithfulness [43] randomization [27]
Faithfulness (Correlation)	Bhatt et al.	2020	[36]	faithfulness [27,36] fidelity-soundness [30]
Faithfulness Estimate	Alvarez-Melis and Jaakkola	2018	[25]	faithfulness [25,27,40] faithfulness-fidelity-soundness [30]
Feature Agreement (FA)	Krishna et al.	2022	[44]	faithfulness (ground truth) [45]

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3.15.1. *Effective complexity*

The concept of effective complexity, as introduced by Nguyen and Martínez (2020), serves as a metric for evaluating the interpretability of machine learning models. This metric quantitatively assesses the significance of feature attributions by identifying the minimum number of features that, when considered in isolation, have a collective impact on the model's output that is within a predefined tolerance level. The essence of this approach lies in its ability to measure the extent to which

Table 1 (*continued*).

Name of the metric	Author(s)	Year	Source	What does it measure?
Feature mutual information	Nguyen et al.	2020	[42]	simplicity, broadness [42] interpretability-broadness, simplicity [30]
FeatureOverlap	Lakkaraju et al.	2017	[23]	interpretability-simplicity [30]
Focus	Arias-Duart et al.	2022	[46]	faithfulness [46] localization [27]
Fraction of classes	Lakkaraju et al.	2016	[33]	fidelity-completeness [30]
Fraction uncovered	Lakkaraju et al.	2016	[33]	interpretability-clarity [30]
High Faithfulness	Bhatt et al.	2020	[36]	faithfulness [36]
Implementation invariance	Sundarajan et al.	2017	[16]	fidelity-soundness [30]
Infidelity	Yeh, et al.	2019	[34]	fidelity [34] faithfulness [27,30]
Input variance	Kindermans et al.	2019	[47]	fidelity-soundness [30]
Iterative Removal Of Features (IROF)	Rieger et al.	2020	[48]	interpretability [48] faithfulness [27]
Justification score	Laugel et al.	2019	[32]	ground-truth justification [32] fidelity-soundness [30]
Local Lipschitz Estimate	Alvarez-Melis and Jaakkola	2018	[25]	robustness [27] fidelity -soundness [30]
Location instability	Zhang et al.	2018	[49]	instability [49] interpretability-clarity [30]
(Low) complexity	Bhatt et al.	2020	[36]	complexity [36]
(Low) sensitivity	Bhatt et al.	2020	[36]	sensitivity [36]
Max width	Lakkaraju et al.	2017	[23]	interpretability size [23] interpretability-simplicity [30]
Max-Sensitivity	Yeh et al.	2019	[34]	fidelity [34] robustness [27]
Monotonicity metric	Nguyen et al.	2020	[42]	faithfulness [27] soundness [21] fidelity-soundness [30]
Monotonicity metric	Arya et al.	2019	[50]	"goodness" [50] faithfulness [27]
MPRT (Model Parameter Randomization Test)	Adebayo et al.	2018	[51]	randomization [27]
Neighbourhood-fidelity (NF)	Plumb et al.	2019	[52]	fidelity, stability [52] fidelity-soundness [30]
Network Dissection	Bau et al.	2017	[53]	interpretability [53] interpretability-clarity [30]
No. of descriptors	Lakkaraju et al.	2017	[23]	interpretability-simplicity [30]

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interactions between features contribute to the model's behaviour. Unlike traditional metrics that evaluate features independently, effective complexity acknowledges the interplay among features, thereby offering a more holistic view of model interpretability [42].

A model with low effective complexity is characterized by its ability to maintain predictive accuracy even when a number of theoretically significant features are disregarded. This attribute suggests that the model can afford to overlook certain features without substantial loss

Table 1 (*continued*).

Name of the metric	Author(s)	Year	Source	What does it measure?
No. of predicates	Lakkaraju et al.	2017	[23]	interpretability -simplicity [30]
Non-representativeness	Nguyen et al.	2020	[42]	interpretability - parsimony, simplicity [21] fidelity-completeness [30]
Non-sensitivity	Nguyen et al.	2020	[42]	fidelity-soundness [30] axiomatic [27]
Non-sensitivity:	Ylikoski, P., Kuorikoski, J.	2010	[54]	fidelity-soundness [30] soundness [21]
Pairwise Rank Agreement (PRA)	Krishna et al.	2022	[44]	disagreement [44] faithfulness (ground truth) [45]
Pearson correlation	Adebayo et al.	2018	[51]	fidelity-soundness [30]
Pixel Flipping	Bach et al.	2015	[55]	faithfulness [27]
Prediction Gap on Important feature perturbation (PGI)	Petsiuk et al. i Dai et al.	2018	[56]	faithfulness (predicted) [45]
Prediction Gap on Unimportant feature perturbation (PGU)	Petsiuk et al. i Dai et al.	2018	[56]	faithfulness (predicted) [45]
Random Logit Test	Sixt et al.	2020	[57]	randomization [27]
Rank Agreement (RA)	Krishna et al.	2022	[44]	faithfulness (ground truth) [45]
Rank Correlation (RC)	Krishna et al.	2022	[44]	faithfulness (ground truth) [45]
Recall of important features	Ribeiro et al.	2016	[35]	fidelity-soundness [30]
Region perturbation (via MoRF (Most Relevant First))	Samek et al.	2015	[58]	faithfulness [27] fidelity-soundness [30]
Relative Input Stability (RIS)	Agarwal, et al.	2022	[26]	robustness [27] stability [26,45]
Relative Output Stability (ROS)	Agarwal, et al.	2022	[26]	robustness [27] stability [26,45]
Relative Representation Stability (RRS)	Agarwal, et al.	2022	[26]	robustness [27] stability [26,45]
Relevance Mass Accuracy	Arras et al.	2021	[59]	localization [27]
Relevance Rank Accuracy	Arras et al.	2021	[59]	localization [27]
ROAD (RemOve And Debias)	Rong, Leemann, et al.	2022	[60]	fidelity [60] faithfulness [27]
ROAR (Remove and Retrain)	Hooker et al.	2018	[61]	fidelity-soundness [21,30]
Robustness	Plumb et al.	2019	[52]	fidelity-soundness [30]
Rule overlap	Lakkaraju et al.	2017	[23]	interpretability-clarity [30]
Runtime operation counts	Slack et al.	2019	[62]	simulatability [62] interpretability-parsimony/ simplicity [21] interpretability-simplicity [30]
Selectivity	Montavon et al.	2018	[63]	faithfulness [27] fidelity-soundness [21,30]

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Table 1 (*continued*).

Name of the metric	Author(s)	Year	Source	What does it measure?
Sensitivity	Sundararajan et al.	2017	[16]	sensitivity [64] fidelity-soundness [30]
Sensitivity	Bhatt et al.	2020	[36]	sensitivity [36] fidelity-soundness [30]
Sensitivity-n	Ancona et al.	2017	[65]	sensitivity [65] faithfulness [27] fidelity-soundness [21]
Sign Agreement (SA)	Krishna et al.	2022	[44]	faithfulness (ground truth) [45]
Signed Rank Agreement (SRA)	Krishna et al.	2022	[44]	faithfulness (ground truth)[45]
Size	Lakkaraju et al.	2017	[23]	interpretability -simplicity [30]
Smooth MPRT	Hedström et al.	2023	[43]	sensitivity [43] randomization [27]
Sparseness	Chalasani et al.	2020	[66]	complexity [27]
Spearman rank correlation	Adebayo et al.	2018	[51]	fidelity-soundness [30]
SSIM	Adebayo et al.	2018	[51]	fidelity-soundness [30]
Sufficiency	Dasgupta et al.	2022	[37]	faithfulness [27,37]
Target Mutual Information	Nguyen et al.	2020	[42]	fidelity-soundness [30] interpretability-broadness [21]
TCAV score (TCAVq)	Kim et al.	2018	[67]	sensitivity [67] fidelity -soundness [30]
Top-K Intersection	Theiner et al.	2021	[68]	localization [27]
Tree ensembles complexity	Hara et al.	2016	[69]	interpretability -simplicity [30]
Trustworthiness	Ribeiro et al.	2016	[35]	fidelity -soundness [30]
Disparity	Dai et al.	2022	[70]	disparity for any other metric

of performance, pointing towards a reduced cognitive load for the users interpreting the model. However, the authors remark that this simplification may come at the cost of factual precision. The trade-off, in this case, promotes a broader and more accessible explanation, emphasizing the importance of simplicity and general applicability in the model's explanations [42].

The metric is reflected in the following equation:

$$k^* = \arg \min_{k \in \{1,\ldots,N\}} |M_k| \quad \text{s.t.} \quad \mathbb{E}\left(l(y^*, f_{M_k}) \mid \mathbf{x}^*_{M_k}\right) < \epsilon$$

where

- Effective complexity (k^*) Identified as the smallest number of top features (k) for which the model, when excluding these features, does not deviate from expected outcomes by more than ϵ
- ϵ A predefined tolerance level, greater than 0, that indicates the acceptable threshold for model deviation.
- $x^{(i)}$ denotes the features associated with each attribution
- M_k Refers to the set comprising the top k features, specifically the ones ranked from $x(N-k)$
- $f-_{M_k}$ The model f adjusted to ignore the top k features identified in M_k, focusing instead on the remaining features.

3.15.2. Completeness

The completeness metric, explained by Sundararajan et al. (2017) serves as a critical evaluative framework for understanding the attribution of input features in deep learning models. This metric is grounded in the principle that the aggregate of attributions assigned to individual input features should equate to the change in the model's output relative to a specified baseline. The essence of this approach is to ensure that the attribution method fully accounts for the influence of input features on the model's predictions, providing a comprehensive and numerically balanced explanation of the model's behaviour [16].

Completeness is anchored in the axiom that the sum of integrated gradients, representing the attribution for each input feature, mirrors the difference in the model's output between the given input and a chosen baseline. This baseline acts as a reference point, often selected such that the model's prediction approximates zero, enabling a straightforward interpretation of how input features contribute to the deviation from this baseline. This concept finds its mathematical underpinning in the fundamental theorem of calculus, specifically adapted for path integrals in the context of deep neural networks.

According to the authors of the metric, the axiom of completeness not only serves as a "sanity check", ensuring that the attribution method offers a holistic view of the input features' contributions, but also establishes a desirable property for models used in scenarios where precise numerical outcomes are critical, such as in calculating insurance premiums based on individual credit features. The metric

implies that a well-designed attribution method should provide a complete and balanced distribution of the model's output across the input features [16].

The metric is expressed by the following equation. Given a function $F : \mathbb{R}^n \to \mathbb{R}$ that is differentiable almost everywhere, the completeness property of Integrated Gradients can be expressed as:

$$\sum_{i=1}^{n} \text{IntegratedGrads}_i(x) = F(x) - F(x')$$

where:

- n: The number of features.
- x: The input vector for which the prediction is evaluated.
- x': A baseline input vector selected such that $F(x') \approx 0$. For instance, in image models, a black image can serve this purpose effectively.
- $F(x)$: The output of the model F for the input x.
- $F(x')$: The model's output for the baseline input x'.
- $\text{IntegratedGrads}_i(x)$: Attributions to the ith input feature calculated using Integrated Gradients.

For many deep networks, it is possible to select a baseline (x') such that the prediction at the baseline is nearly zero. In scenarios involving image models, for example, a black image often serves as an effective baseline [16].

It is noteworthy that this metric was further expanded upon under various nomenclatures such as Summation to Delta by Shrikumar et al. (2017) [72], sensitivity-n by Ancona et al. (2018) [65], and Conservation by Montavon et al. (2018) [38].

3.15.3. Sensitivity-n

The aforementioned sensitivity-n metric, as described by Ancona et al. (2018), plays an important role in assessing the correlation between the aggregated attributions of input features and the consequent variations in the model's output. This metric is predicated on the notion that the sum of attributions assigned to any subset of features, with cardinality n, should mirror the change in output resulting from the omission of these features. In essence, sensitivity-n measures the extent to which the model's output is sensitive to the presence or absence of specific combinations of input features, averaged across multiple test samples for robustness [65]. Mathematically, sensitivity-n is validated when, for all possible subsets of features $x_S = [x_1, \ldots, x_n] \subseteq x$, the cumulative attribution of these features equates to the differential in output caused by their exclusion, symbolically represented as

$$\sum_{i=1}^{n} R_i^c(x) = S_c(x) - S_c\left(x_{[x_S=0]}\right)$$

In the context of this metric, when $n = N$, with N denoting the total number of input features, the equation converges to the principle of completeness or Summation to Delta, thereby positioning sensitivity-n as a generalization of these concepts [65].

To empirically measure sensitivity-n, the approach involves estimating the correlation between the cumulative attributions and output variations for a number of feature subsets of varying cardinalities. This correlation, quantified using the Pearson correlation coefficient (PCC), is calculated by randomly sampling one hundred subsets of features for different values of n, thereby spanning a broad spectrum of potential feature combinations.

3.15.4. Faithfulness estimate

The faithfulness estimate, as conceptualized by Alvarez-Melis et al. in 2018, is an important metric designed to thoroughly evaluate the accuracy and reliability of relevance scores assigned to features by an explanation model. This metric fundamentally assesses the alignment between the predicted importance of features and their actual impact on the model's output. Given the challenge of directly accessing the "true" influence of each feature, the faithfulness estimate adopts a practical approach by focusing on the observable consequences of feature manipulation on model predictions. This involves selectively obscuring or eliminating features and observing the resultant changes in output probability for classification tasks, thereby providing a quantifiable measure of each feature's relevance [25].

The metric employs a systematic process to correlate these observed probability changes with the relevance scores predicted by various interpretability methods, such as LIME, SHAP, and SENN. By conducting experiments across multiple points and compiling aggregate statistics, the methodology allows for a detailed comparison of how different explanation models perform in capturing the true significance of features. This evaluation is particularly insightful when applied to models with an additive structure, where features can be directly manipulated by adjusting their coefficients, offering a more precise and controlled assessment of feature relevance [25].

It is worth mentioning that the faithfulness estimate extends beyond traditional input features to encompass learned concepts within models, highlighting its versatility and applicability across a range of scenarios. This approach not only reinforces the importance of feature selection and manipulation in understanding model behaviour but also underscores the metric's capacity to provide a robust framework for validating the integrity of explanations generated by AI models. As noted by the authors, through its focus on the correlation between theoretical attributions and practical impact, the faithfulness estimate serves as a critical tool in the quest for transparent and accountable AI, ensuring that features selected to be "relevant" by explanation models genuinely reflect their influence on model predictions [25].

3.15.5. Monotonicity metric

The monotonicity metric, as introduced by Luss et al. (2019) [73] and further elaborated upon by Arya et al. in 2019 [50], offers a structured approach to evaluating the incremental impact of features on a model's decision-making process. This metric is fundamentally designed to investigate whether incorporating additional positive evidence (features with positive relevance) systematically enhances the probability of a specific classification outcome. It initiates with a baseline vector, presumed to represent a neutral or non-informative state, and progressively integrates features according to their attributed relevance, prioritizing the inclusion of features in ascending order of their importance [50].

The core hypothesis underpinning the monotonicity metric is that if the features are indeed positively correlated with the targeted decision or classification, then logically, the incorporation of these features should lead to a consistent increase in the classification probability for the relevant class. This sequential addition of features, observed through the lens of their contribution to enhancing classification certainty, provides a detailed insight into the model's interpretability and fairness. By carefully tracking the variation in classification probabilities with each incremental introduction of relevant features, the monotonicity metric not only validates the positive correlation between key features and outcomes but also underscores the importance of feature selection and ordering in explainable AI models [50].

3.15.6. Monotonicity metric

The monotonicity metric proposed by Nguyen and Martínez in 2020 offers a methodology for evaluating the relationship between feature attributions and the associated uncertainty in model predictions. This metric employs the Spearman's rank correlation coefficient (ρ_S) to analyse the monotonic relationship between the absolute values of feature attributions and the estimated expectations of model uncertainty for each feature. Essentially, this approach constructs two vectors: one (a) containing the absolute values of the attributions assigned to each feature, and the other (e) encapsulating the corresponding expectations of uncertainty, as derived from the model's probability estimations [42].

As a result, the equation for the monotonicity for feature attributions a_i is $\rho_S(\mathbf{a}, \mathbf{e})$ [42].

By focusing on the Spearman's rank correlation, this metric transcends mere numerical comparisons, instead assessing the rank-based relationship between the magnitude of a feature's attribution and its contribution to the model's uncertainty. A key premise of this metric is that a meaningful attribution should display a monotonic relationship with the model's uncertainty estimates; in other words, features seen as more important should consistently correlate with greater impacts on the uncertainty of the model's predictions.

A non-monotonic correlation, as flagged by this metric, suggests that the attributions may not accurately reflect the true significance of the features, thereby calling into question the reliability of the attribution method in conveying the features' importance. This metric thus serves as a critical tool in the evaluation of attribution methods, ensuring that the identified feature importances are not only statistically significant but also logically consistent with the model's predictive behaviour and uncertainties [42].

As noticed by Nguyen and Martínez, their proposed methodology significantly differs from a similar measure proposed by Arya et al. (2019), which also explores monotonicity but does so in relation to a probability drop rather than the uncertainty in probability estimation as in Nguyen and Martínez's approach. Moreover, Nguyen and Martínez's metric introduces the flexibility to adapt to regression tasks depending on the loss function (l) selected. This distinction underscores the adaptability and broad applicability of Nguyen and Martínez's metric in evaluating the relationship between feature importance and model uncertainty across different model types and tasks [42].

3.15.7. Selectivity

The Selectivity metric, introduced by Montavon et al. in 2017, represents a critical approach to evaluating the quality of explanations generated by predictive models. This metric gauges the immediate impact on the prediction function's output as the most relevant features, according to the attribution scores, are sequentially removed. Originally conceptualized for image data through a technique known as "pixel-flipping" and later adapted to textual data, where high-relevance words are effectively nullified by setting their embeddings to zero, this method provides a direct measure of the attribution's accuracy and relevance. The underlying rationale of the Selectivity metric is straightforward yet powerful: it assumes that if the removal of features deemed most important by the model results in a significant and rapid decline in the prediction function's output ($f(x)$), then the attribution method has successfully identified the truly impactful variables. The steepness of the output's drop is quantitatively assessed using the Area Under the Curve (AUC) metric, where a lower AUC score signifies a sharper decline, hence affirming the high relevance of the removed features [63].

This metric emphasizes the importance of precision in the distribution of relevance scores across the features, advocating for a redistribution that highlights those with the most substantial influence on the model's function. By averaging the AUC results across a broad spectrum of examples within a dataset, the Selectivity metric offers a robust means to validate the fidelity of explanations provided by attribution methods, ensuring that they accurately reflect the contributions of features to the predictive outcomes [63].

3.15.8. Sufficiency

The sufficiency metric, as introduced by Dasgupta et al. in 2022, offers a comprehensive framework for assessing the reliability and relevance of explanations provided by predictive models. This metric is founded on the principle that if a particular explanation or property (π) justifies a prediction for one instance (x), then any other instance (x') possessing the same property should logically receive the same prediction, irrespective of potentially differing explanations [37].

According to the authors' reasoning, illustratively, if an image x is explained as "contains a zebra" leading to a prediction label $f(x) = savannah$, then another distinct image x' that also features a zebra should be assigned the same label, even if its explanation differs, say, "contains a baobab tree". This premise ensures that the explanation framework not only provides intelligible and consistent justifications across instances but also respects the intrinsic significance of key properties or features that fundamentally determine the prediction [37].

To operationalize the metric, the concept of intelligibility is introduced, positing that it should be feasible for humans to verify whether a given property π is applicable to an instance x. This leads to the formulation of a relation $A(x, \pi)$, allowing the identification of a set Cx comprising instances that share the same explanatory property as x. Sufficiency is then quantitatively assessed through the homogeneity of predictions across Cx, employing a probability distribution over this set to calculate the likelihood that any instance within Cx will be classified identically to x. In other words, the metric can be expressed with the following equation: $C_x = \{x' \in \chi : A(x', e(x))\}$ [37].

Defined at both local and global levels, sufficiency offers a measure of an explanation system's effectiveness. Local sufficiency evaluates the predictability and coherence of explanations for individual instances, while global sufficiency aggregates this assessment across the entire model, providing a broader perspective on the explanatory power and reliability of the model's justifications. This dual approach enables a comprehensive evaluation of how well explanations align with the model's predictive outcomes, reinforcing the importance of meaningful and consistent explanations in fostering trust and understanding in AI systems [37].

3.15.9. MPRT

The Model Parameter Randomization Test (MPRT), proposed by Adebayo et al. in 2018, introduces a novel framework for evaluating the sensitivity of saliency methods to the underlying parameters of a model. This test is predicated on the hypothesis that the explanations generated by saliency methods should fundamentally depend on the model's learned parameters, which captures the insights gained from the data during training. By comparing the explanations derived from a fully trained model against those generated from a version of the model with randomized parameters, the MPRT aims to work out the degree to which a saliency method is influenced by the model's parameterization [51].

The essence of the MPRT lies in its approach to parameter randomization, which can be implemented in two distinct manners: a comprehensive randomization of all model weights, and a more targeted randomization affecting single layers independently while maintaining the original parameters in all other layers. These variations allow for a comprehensive assessment of a saliency method's responsiveness to changes in the model's structure and parameter settings [51].

A key metric for the MPRT is the distance (or discrepancy) between the explanations generated under the original and randomized settings. A significant discrepancy in explanations would indicate a strong dependence on the model's parameters, validating the utility of the saliency method for tasks that require insight into the model's functioning, such as debugging. Conversely, similarity between explanations across both settings would suggest that the saliency method is not effectively capturing the model's learned behaviours, rendering it less useful for understanding or improving the model's performance. Through this precise evaluation, the MPRT provides valuable insights into the reliability and effectiveness of saliency methods in revealing the contributions of model parameters to the decision-making process [51].

3.15.10. RIS

The Relative Input Stability (RIS) metric, introduced by Agarwal et al. in 2022, advances the examination of explanation methods by measuring the proportional disparity between explanations for pairs of input instances relative to the difference in the inputs themselves. This metric innovatively frames the evaluation of explanations within the context of input perturbations, thereby offering a perspective on the stability and reliability of explanatory outcomes. The metric is expressed using the following equation: $RIS(x, x', \mathbf{e}_x, \mathbf{e}_{x'}) = max_{x'} \frac{\left\| \frac{e_x - e_{x'}}{e_x} \right\|_p}{max(\left\| \frac{x - x'}{x} \right\|_{p, \epsilon_{min}})}, \forall x' s.t. x' \in \mathcal{N}x; \hat{y}_x = \hat{y}_{x'}$ [26]. Central to RIS is the calculation of the relative distance between explanations $\mathbf{e}_x$ and $\mathbf{e}_{x'}$ for two inputs x and x', standardized by the distance between the inputs. This is quantified as the ratio of the change in explanations to the change in inputs, ensuring that the comparison remains meaningful across a spectrum of attribution methods with inherently varied scales. A key aspect of RIS is its incorporation of a normalization factor (ϵ_{min}) to mitigate potential division by zero, thereby enhancing the metric's robustness.

The foundational premise of RIS posits that inputs which are closely related should yield explanations that are similarly close, indicating a direct correlation between the similarity of inputs and the congruence of their explanations. A higher RIS value signals greater instability in the explanation relative to the input perturbation, suggesting that the explanation method may vary significantly even for minor changes in input. This insight is critical for assessing the fidelity of explanations, especially in contexts where understanding model behaviour in the face of input variations is paramount. RIS thereby serves as a critical tool for evaluating the coherence and consistency of explanations across different instances, ensuring that explanation methods not only provide insightful interpretations but also maintain a stable and reliable relationship with the inputs they aim to explain [26].

3.15.11. Attribution localization

The attribution localization metric, proposed by Kohlbrenner et al. in 2019, introduces a quantitative framework for evaluating the effectiveness of attribution methods in the context of object detection and classification tasks. This metric focuses on assessing the extent to which the positive attributions – those contributions identified as relevant for the decision-making process of a model – are accurately localized within the bounds of the targeted object in an image. By comparing the proportion of relevance attributed to the object against the total positive relevance across the entire image, this measure provides insight into the precision with which an attribution method can identify and highlight the actual object of interest [29].

The metric is defined in two parts: the inside-total relevance ratio (μ) and a weighted variant (μ_w). The former, (μ), calculates the ratio of the sum of positive attributions within the object's bounding box (R_{in}) to the total sum of positive attributions in the image (R_{tot}): $\mu = \frac{R_{in}}{R_{tot}}$. The weighted variant, μ_w, further refines this assessment by considering the relative size of the object to the image; it multiplies μ by the ratio of the total image area (S_{tot}) to the object area (S_{in}), thereby normalizing the measure to account for object size: $\mu_w = \mu \cdot \frac{S_{tot}}{S_{in}}$ This comprehensive approach allows μ_w to place additional emphasis on the correct localization of smaller objects, recognizing the increased challenge of accurately attributing relevance to objects that occupy a smaller portion of the image space. Higher values of μ and μ_w indicate a greater concentration of relevant attributions within the object area, signifying a higher degree of accuracy in pinpointing the elements critical for the model's decision [29].

By sidestepping numerical complications related to bounding box sizes, these measures offer a robust and scalable criterion for evaluating and comparing the localization capabilities of different attribution methods. Through this, the attribution localization metric serves as an important tool for measuring the precision and reliability of explanations provided by object detection and classification models, ensuring that attributions meaningfully reflect the visual features that inform the model's predictions [29].

3.16. xAI evaluation taxonomies and frameworks

This section provides answers to the last Research Question (RQ3), "What taxonomies or frameworks exist for evaluating xAI systems, and how do they categorize and measure explanation quality?"

Having outlined a number of explanation properties which can be measured, it is worth mentioning again that there is no scientific consensus among the researchers, who have proposed various xAI evaluation taxonomies in the literature, reflecting the multifaceted nature of explainability and interpretability.

3.16.1. xAI evaluation taxonomies

Zhou et al. review quantitative metrics for assessing the quality of explanation methods and aligns with Lopes in listing properties of explainability, including interpretability (which encompasses clarity, broadness, parsimony a.k.a. simplicity), and fidelity (consisting of completeness, soundness) [21,30]. Similarly, Markus et al. divide explainability into two main components: interpretability (clarity and parsimony) and fidelity (completeness and soundness), focusing on the clarity and truthfulness of explanations as well as their comprehensiveness and accuracy [24].

Alvarez-Melis and Jaakkola focus on the aspects of explicitness/intelligibility, faithfulness, and stability, providing a view on how explanations can be evaluated based on their clarity, accuracy to the model's decisions, and consistency across similar inputs [25].

Guidotti et al. expand the evaluation criteria to encompass interpretability/comprehensibility, accuracy, and fidelity, alongside considerations such as fairness, privacy, monotonicity, usability, reliability, robustness, causality, scalability, and generality. Guidotti et al. also highlight the often-overlooked aspect of model complexity, emphasizing its inverse relationship with Interpretability [22].

Molnar introduces a distinction between objective evaluations and human-centred evaluations, focusing on aspects like sparsity, interaction strength, fidelity, sensitivity to perturbations, and simulatability [17]. Meanwhile, Lakkaraju et al. emphasize fidelity, unambiguity, interpretability, interactivity, providing formulas for a more quantifiable approach, albeit with a nod towards human-based evaluation [23].

Lastly, Nauta et al. propose a distinct taxonomy addressing correctness, completeness, consistency, continuity, contrastivity, covariate complexity, compactness, composition, confidence, context, coherence, and controllability, offering a comprehensive framework for assessing explanations [28].

Upon analysing the various xAI evaluation taxonomies, we can observe both similarities/overlaps and distinct differences among the proposed frameworks. These taxonomies reflect the multifaceted nature of explainability and interpretability in AI systems, illustrating the breadth of criteria from the technical aspects of explanations. Although each taxonomy offers unique insights into different dimensions of explainability, reflecting the interdisciplinary nature of XAI evaluation and the ongoing dialogue on how best to measure the effectiveness of explanatory models, it is evident that there is no singular, universally accepted framework. This diversity signifies both the richness and the complexity of the field, where overlapping categories and sometimes overly intricate classifications underscore the challenges of achieving consensus. The multiplicity of perspectives not only illustrates the depth of the domain but also highlights the necessity for continued research and dialogue to refine and possibly unify these frameworks for broader applicability and understanding.

3.16.2. xAI evaluation frameworks

The demand for accountable and interpretable AI systems grows has led to the development of various frameworks aimed at measuring the quality of explainability in machine learning systems. Some examples of such initiatives are Quantus, OpenxAI, and AI360, each offering open and free resources to deepen users' understanding and measurement of AI explainability. These frameworks not only democratize access to

explainability metrics but also enrich the discourse on ethical AI by providing diverse measures of interpretability.

Quantus [27] is notable for its extensive collection of metrics, offering over thirty different measures that enable users to evaluate various properties of explainability, e.g. faithfulness, robustness, complexity, etc. across multiple AI models and datasets. This wide range of metrics ensures that researchers and practitioners can conduct thorough analyses tailored to the specific characteristics and requirements of their AI systems. Quantus's emphasis on a comprehensive toolkit aids in dissecting the details of explanation quality, making it a valuable asset for those looking to assess and enhance the transparency of their models.

The metrics included in the framework fall into six categories and are as follows:

1. Faithfulness: faithfulness correlation, faithfulness estimate, monotonicity metric (Arya et al. 2019), monotonicity metric (Nguyen et al. 2020), pixel flipping, region perturbation, selectivity, sensitivity-n, IROF, infidelity, ROAD, sufficiency.
2. Robustness: local Lipschitz estimate, max-sensitivity, avg-sensitivity, continuity, consistency, Relative Input Stability (RIS), Relative Representation Stability (RRS), Relative Output Stability (ROS).
3. localization: pointing game, attribution localization, top-K intersection, relevance rank accuracy, relevance mass accuracy, AUC, Focus.
4. Complexity: sparseness, complexity, effective complexity.
5. Randomization (Sensitivity): MPRT, smooth MPRT, efficient MPRT, random logit test.
6. Axiomatic: completeness, non-sensitivity, input invariance [27].

OpenXAI [45], on the other hand, focuses on fostering a community-driven approach to the exploration of explainable AI. While it includes a broad spectrum of metrics, its strength lies in the collaborative environment it cultivates for the development, sharing, and refinement of explainability methods. By encouraging open contributions, OpenXAI accelerates the innovation of explainability techniques and their evaluation, facilitating a dynamic exchange of ideas that propels the field forward. The framework includes the following metrics:

- Ground-truth Faithfulness: Feature Agreement (FA), Rank Agreement (RA), Sign Agreement (SA), Signed Rank Agreement (SRA), Rank Correlation (RC), Pairwise Rank Agreement (PRA).
- Predicted Faithfulness: Prediction Gap on Important feature perturbation (PGI), Prediction Gap on Unimportant feature perturbation (PGU).
- Stability: Relative Input Stability (RIS), Relative Representation Stability (RRS), Relative Output Stability (ROS).
- Fairness [45].

In turn, AI360 [40], developed by IBM, is a pioneer in integrating fairness alongside explainability, offering a suite of tools that assess not only the interpretability of AI systems but also their fairness. With a focus on the ethical dimensions of AI, AI360 equips users with the ability to evaluate models on both fronts, providing insights into how systems can be made both more understandable and equitable. It includes two evaluation metrics, faithfulness (Alvarez-Melis and Jaakkola, 2018) and monotonicity (Luss et al. 2019) [40].

4. The experimental part: materials and methods

This section presents the methodology and the results of the experimental part of the study presented. The practical part of this work consisted in two experiments in which researchers working independently first applied xAI techniques to the distinct AI/ML models they used in their works, and then subsequently used the evaluation metrics included in the conventional xAI evaluation frameworks described above, in order to assess the explanations provided to them.

In the following subsections, the data collection processes are described, along with the models deployed, xAI techniques used, and the evaluation metrics that the researchers managed to successfully apply. The results and insights from these experiments are discussed in the next Section.

4.1. Data collection and preprocessing

The first experiment employs the CIC IoT dataset 2023 [74], a real-time dataset and benchmark for large-scale attacks in Internet of Things (IoT) environments. This dataset comprises 33 attacks categorized into seven types, including Denial of Service (DoS), and Distributed DoS (DDoS), Recon, Web-based, Brute Force, Spoofing, and Mirai, targeting 105 devices in an IoT topology. The choice to select this dataset was dictated by various key factors, such as its novelty. Additionally, this dataset is based on real-world data, reflecting the actual threats and challenges faced in IoT environment. It provides a comprehensive coverage of possible threats faced by IoT devices and, due to the dataset's diverse range of attacks and devices employed, it has a general applicability to a wide scope within Network Intrusion Detection. The preprocessing steps included removing missing values and duplicate records. For the dimensionality reduction, the 15 most significant features were selected using SelectKBest from scikit-learn, and label encoding was applied to the target variable. The dataset was split into training and testing sets with a 80:20 ratio, and feature scaling was performed using StandardScaler from scikit-learn.

In the second practical experiment of this study, the "CSE-CIC-IDS2018 Intrusion Detection Evaluation Dataset"[1] was used (commonly referred to as the IDS 2018 Intrusion CSVs), accessible on Kaggle. This dataset is a comprehensive collection of network traffic and intrusion detection events, compiled to aid in the development and evaluation of intrusion detection systems. A significant component of this dataset focuses on the analysis of Distributed Denial of Service attacks, making it particularly valuable for the research into the effectiveness of various detection methodologies against such threats. The choice of this dataset was driven by its richness in real-world attack scenarios, including a wide range of attack vectors, which provides a solid foundation for evaluating the performance of the tested models.

4.2. The models employed

In the first part of the experiment, a neural network (NN) model was implemented through PyTorch to perform intrusion detection tasks on the selected dataset. The NN model consists of multiple layers, including fully connected (dense) layers followed by activation functions. Specifically, a two-layer architecture with Rectified Linear Unit (ReLU) activation functions between layers to introduce non-linearity into the model. The input layer of the model receives the preprocessed feature vectors extracted from the dataset. These feature vectors represent various attributes and characteristics of devices and network traffic. The output layer of the model produces predictions regarding the presence or absence of attacks based on the input data. The model was then trained using the Adam optimizer with a learning rate of 0.01. The Cross-Entropy Loss function was used as the optimization criterion, suitable for classification tasks. During training, the model parameters were adjusted to minimize the discrepancy between predicted and actual attack labels in the training data. To assess the performance of the model, standard evaluation metrics such as accuracy, precision, recall, and F1-score were employed.

In the second part of the experiment, three distinct machine learning models were employed, each chosen for their unique characteristics and compatibility with the dataset. These models, implemented using

[1] https://www.kaggle.com/datasets/solarmainframe/ids-intrusion-csv?resource =download

TensorFlow, were specifically selected to ensure a diverse approach to the analysis. The models include the RandomForestClassifier and CartModel, both run on a Linux environment, and the Sequential model, executed on Windows. Each model was trained on a sample of 200,000 records from the previously mentioned CSE-CIC-IDS2018 dataset, aimed at providing a comprehensive understanding of their performance in detecting DDoS attacks.

The selection of these models was driven by a desire to incorporate a range of analytical techniques within the study. Initially, there was an interest in utilizing scikit-learn models to further diversify the analysis approach. However, compatibility issues with the necessitated the use of models developed in either TensorFlow or PyTorch frameworks, led to the exclusive selection of TensorFlow models.

4.3. Selection and implementation of xAI techniques

In the first part of the experiment, the xAI techniques selected were implemented using the Captum library. Specifically, Integrated Gradients (IG) and SHapley Additive exPlanations (SHAP) [75] were employed to interpret the model's predictions. IG is a well-established method for unpacking how a Deep Neural Network (DNN) arrives at its predictions by analysing the influence of each input feature. It computes the integral of the gradients along a straight path, starting from a baseline input. On the other hand, SHAP is based on cooperative game theory principles. It assigns each feature an importance value by considering its contribution to the difference between the actual model output and the expected output. The motivation for selecting these explainers lies in their ability to offer a comprehensive understanding of the model predictions. These techniques provide complementary perspectives on feature importance and model behaviour. What is more, these explainers are widely adopted methods in the field of xAI, thanks to their trustworthiness and versatility.

In the second part of the experiment, three acknowledged xAI techniques were implemented, namely SHAP, LIME, and ANCHORS. The selection of these techniques was primarily influenced by their compatibility with the Quantus xAI evaluation framework. These techniques were instrumental in extracting the necessary data insights required by Quantus, as well as facilitating interpretability across other analytical frameworks. The utilization of SHAP, LIME, and ANCHORS allowed the authors to gather critical insights into the decision-making processes of the applied models.

However, it is crucial to note the unique challenge that was encountered with the OpenXAI framework, where the ANCHORS technique could not be incorporated. Despite its effectiveness and compatibility with Quantus, ANCHORS was not natively supported by OpenXAI, and the attempts to integrate this technique were unsuccessful.

4.4. Evaluation metrics

In the first part of the experiment, for evaluating the xAI techniques, a subset of the metrics from the Quantus library have been selected. These metrics assess different aspects of xAI explanations, including randomization, complexity, faithfulness, robustness, localization, and axiomatic properties. The metrics were selected based on their relevance to the presented study and suitability for tabular data. In particular, the randomization metric tests how much explanations deteriorate as the inputs to the evaluation problem change. For the present analysis, the Model Parameter Randomization Test (MPRT) from [76] has been used. The MPRT randomly shuffles the setting of one layer of the model, checks how much the explanations change compared to the original one, and then repeat these steps for multiple layers to evaluate the overall effect on the explanations. Complexity refers to the conciseness of the explanations, which means that a limited number of features is used to explain the model prediction. To evaluate the complexity within the experiments, the complexity metric defined in [77] and the effective complexity from [42] have been selected. Within the xAI metrics, faithfulness quantifies how explanations follow the predictive behaviour of the model. Among the others, the sufficiency metric defined in [78], that measures the extent to which similar explanations have the same prediction label, has been employed to evaluate faithfulness. Additionally, the monotonicity metric [42] has been evaluated as well. Robustness measures how stable the explanations are when subject to minor perturbations of the input, under the assumption that the model has not been altered. For the experiments, the Relative Input Stability (RIS), Relative Output Stability (ROS), and Relative Representation Stability (RRS) from [79] have been used. The localization metric tests if the explainable evidence is centred around a Region of Interest (ROI). All the possible metrics for this parameter appear to be tailored for image-based datasets, so they were tested but did not yield meaningful results. However, the attribution localization defined in [80], the AUC, and the pointing game from [59] were successfully applied to tabular data. For the axiomatic properties evaluation, the completeness metric defined in [16] has been chosen. This parameter evaluates if the sum of attributions is equal to the difference between the function values at the input x and baseline x'.

Table 2
Summary of xAI metrics for the different explainers using Quantus for NN model.

Metric	IG	SHAP
MPRT	0.9854	0.9873
Complexity	0.8975	0.8946
Effective complexity	5.3860	5.3858
Monotonicity	0.8975	0.8946
Sufficiency	0.7969	0.9550
RIS Agarwal	0.0281	0.0307
ROS Agarwal	10 077	10 091
RRS Agarwal	144 481	144 682

For the second part of the experiment, the selection of metrics was also primarily guided by their compatibility with the models employed. The choice of metrics was fundamentally constrained to those that functioned effectively with the selected models. For the sake of clarity, the results containing negative indicators only were not presented in the tables. In the case of this experiment, the remaining metrics failed to deliver due to inherent conflicts with the model constructions, model types, and the nature of the data being processed. Such challenges underscore the intricate relationship between model architectures and the applicability of specific metrics, highlighting the necessity for careful metric selection to ensure meaningful evaluation.

Within the scope of the OpenXAI framework, the metric selection was further refined to accommodate the framework's unique requirements. For ground truth metrics, the set including ['PRA', 'RC', 'FA', 'RA', 'SA', 'SRA'] was used. The application of these metrics revealed compatibility issues, with errors indicating that they align effectively only with linear models. Furthermore, for stability metrics, ['RIS', 'RRS', 'ROS'] were employed. Again, the exploration into these metrics highlighted a notable compatibility issue.

5. Experimental results

Table 2 summarizes the results of the first part of the experiment, i.e., the evaluation of xAI techniques using the metrics listed in the previous section using IG and SHAP as explainers, in the context of intrusion detection.

Tables 3–5 show the results of the second part of the experiment.

Table 2, summarizing xAI metrics for different explainers (IG and SHAP), presents some insights into their performance. The MPRT scores are exceptionally high for both IG and SHAP, suggesting robustness in perturbation-based relevance attribution, which is crucial for intrusion detection systems where detecting slight variations in input can be critical. The near-parity in scores indicates both methods' effectiveness, with SHAP slightly outperforming IG, possibly due to its more

Table 3
The results for DNN — 200k samples.

DNN 200k samples			
QUANTUS			
Metric	SHAP	LIME	ANCHORS
Complexity	3.0050	3.0242	3.6463
Sparseness	0.8015	0.6878	0.6311
Effective complexity	67.0	69.0	69.0
Completeness	0.0	0.0	0.0
NonSensitivity	2.0	2.0	2.0
Faithfulness Estimate	−0.2970	−0.0225	−0.6216
Pixel-Flipping	0.3329	0.3330	0.3299
Region Segmentation	0.0005	−0.0012	0.0025
Monotonicity-Arya	1.0	1.0	1.0
Monotonicity-Nguyen	0.1682	0.1608	0.2010
Selectivity	0.3329	0.3346	0.3309
Sufficiency	0.0	0.0	0.0
Consistency	0.0	0.0	0.0
AIX360			
Faithfulness	0.2953	0.0219	0.1038
Monotonicity	False	False	False
OpenXAI			
PGU	0.0002	0.0005	–
PGI	0.0004	0.00008	–

Table 4
The results for RF — 200k samples.

RF 200k samples			
QUANTUS			
Metric	SHAP	LIME	ANCHORS
Complexity	2.5674	2.9273	3.6463
Sparseness	0.8698	0.8068	0.6311
Effective complexity	55.0	69.0	69.0
AIX360			
Faithfulness	−0.9066	0.9165	0.1802
Monotonicity	False	False	False
OpenXAI			
PGU	0.2370	0.2326	–
PGI	0.6812	0.6780	–

Table 5
The results for CartModel — 200k samples.

CartModel 200k samples			
QUANTUS			
Metric	SHAP	LIME	ANCHORS
Complexity	1.4241	3.0320	3.6463
Sparseness	0.9577	0.7426	0.6311
Effective complexity	5.0	69.0	69.0
AIX360			
Faithfulness	−0.7967	0.7713	−0.0523
Monotonicity	True	True	True
OpenXAI			
PGU	0.00	0.00	–
PGI	0.9184	0.9999	–

detailed approach to feature contribution analysis. Complexity metrics and Monotonicity yield similar results for IG and SHAP, indicating a close alignment in their interpretability intricacies and consistency in monotonic behaviour across both explainers. This suggests that both methods manage to maintain a balance between interpretability and complexity effectively.

The notable difference in Sufficiency between IG and SHAP highlights SHAP's superior capacity to provide just-enough evidence for predictions, which is particularly valuable in high-stakes environments like intrusion detection. Metrics such as RIS, ROS, and RRS indicate the scalability of both methods, with slight variations that might be due to their inherent differences in handling model outputs. The presence of NaN and False values in some metrics (e.g., Attribution Localization, AUC, Pointing Game, Completeness) suggests limitations in applicability or the challenging nature of these metrics for both explainers within this specific context.

Across the Tables 3, 3, and 5, one can observe the performance of SHAP, LIME, and ANCHORS across various metrics on different models (DNN, RF, CartModel) using 200k samples each. As far as complexity and Sparseness are concerned, SHAP consistently shows lower complexity compared to ANCHORS across all models, suggesting it provides a more straightforward interpretation mechanism. However, ANCHORS appears to produce more sparse explanations, which might be preferable in scenarios where pinpointing the most critical features is essential. In turn, effective complexity demonstrates a significant variance across models, with DNN and CartModel showing extremes in complexity when analysed with SHAP and LIME. This might reflect the differences in how these models process features and the challenges in attributing their decisions to individual inputs. Notably, SHAP tends to have higher Faithfulness scores in the AIX360 context, indicating that its explanations are more aligned with the actual model outputs. The uniformity in Monotonicity results across all models for CartModel suggests a consistent application of this principle, irrespective of the explainer used.

The PGU and PGI metrics, available only under the OpenXAI context for some models, offer a glimpse into the precision of explanations. The absence of values for ANCHORS in several metrics suggests limitations or challenges in using this technique across all contexts.

The detailed examination across different models and xAI techniques reveals a complex relation between the choice of explainer, model architecture, and the specific metric being evaluated. While SHAP and LIME provide a broad consistency in their explanatory power across different models, ANCHORS seems to encounter more limitations, potentially due to its more focused approach to explanation which might not align with all metrics. The discrepancies and challenges observed, such as in Completeness, Attribution Localization, and various stability metrics, underscore the need for ongoing development and refinement of both xAI techniques and the metrics used to evaluate them. This development is necessary for advancing the field of xAI, especially in critical applications like intrusion detection, where the accuracy and interpretability of explanations can significantly impact security outcomes.

The evaluated xAI techniques, especially SHAP, demonstrate a promising ability to explain model decisions across different contexts. However, the varied performance across different metrics and models underscores the complexity of xAI evaluation. It highlights the need for ongoing development to address the gaps in compatibility and comprehensiveness of xAI metrics.

The variations in results, especially when comparing metrics like Faithfulness across different frameworks (e.g., AIX360's faithfulness metric vs. Quantus's faithfulness estimate), reveal discrepancies that might arise from the use of non-native models and explainers within these frameworks. This suggests that while these frameworks offer valuable tools for xAI analysis, their built-in models and explainers are optimized for compatibility, leading to potential challenges when external models are introduced. The variations in results could reflect not only on the inherent differences in the xAI techniques but also on the compatibility and adaptability of the evaluation frameworks to diverse models and data types. This underscores the necessity for flexible and adaptable xAI solutions that can accommodate a wide range of models and data, ensuring meaningful and accurate interpretability.

Finally, it also need to be mentioned that although in both experiments all the metrics available in the three frameworks described in this work have been tested, a fraction of them yielded any results. This may be related to the specificity of the tasks or models used in the study, but also may be indicative of a bigger issue, where the vast majority of metrics do not really serve their function. This is the sign of possible disconnect between the theoretical development of metrics

and their applicability in real-world scenarios. It once again emphasizes the necessity to reevaluate how metrics are designed, validated and applied. Future research should focus not only on developing new metrics but also on critically testing the existing ones, across various tasks and models.

6. Conclusions

The evaluation of explainable AI performed in this study has emphasized the complexity of metrics, frameworks, and practical applicability that define the field. By cataloguing specific measures of explanation as proposed by researchers, this work not only navigates the theoretical foundations of xAI but also progresses into the empirics, evaluating the effectiveness of various xAI methods through experimentation.

The theoretical exploration in the form of a literature survey highlighted a critical insight: the domain of xAI metrics is not only vast but also continuously evolving, marked by a diversity of metrics and an apparent lack of consensus on fundamental properties. This proliferation of metrics, while reflective of the dynamic nature of xAI, poses significant challenges in terms of standardization and comparability. The study's examination of specific metrics highlights the differences in interpretation and application, underscoring the pressing need for a unified framework that would be able to embrace the complexity of explainability.

Experimentally, the study applied these metrics to assess the performance of various xAI techniques. The findings reveal a scenario where techniques like SHAP and IG showcase considerable promise in offering stable and interpretable explanations. However, discrepancies in metric outcomes across different frameworks signal a broader issue of compatibility and adaptability, emphasizing the necessity for xAI solutions that are both flexible and comprehensive. Furthermore, while both theoretical exploration and practical application have dealt with almost a hundred of xAI metrics, the effective yield of actionable insights from these metrics remains modest. This casts a light on the imperative for careful validation and contextual interpretation to ensure the relevance and reliability of the results obtained.

The experimental part of this paper aims to address this gap by providing empirical evidence to evaluate the meaningfulness of these metrics. Theoretical discussions and proposed metrics often assume certain conditions or behaviours that may not hold true in practical scenarios. Experimental validation is crucial to determine if these metrics can reliably measure what they intend to in real-world applications. With a multitude of metrics available, it becomes essential to understand their relative strengths and weaknesses. Experiments allow for direct comparison under controlled conditions, highlighting which metrics provide more actionable insights. In addition, metrics may perform differently depending on the type of data, model, and domain of application. Empirical results help identify context-specific performance, allowing to evaluate the meaningfulness of these metrics

Even the preliminary results indicate noticeable inconsistencies among metrics, even if the metrics seem theoretically sound. Many metrics show high degrees of correlation with each other, suggesting redundancy. This finding is crucial as it implies that the current proliferation of metrics may not add substantial value beyond a certain point.

The study presented bridges the gap between theoretical discourse and empirical validation, presenting an advanced analysis that contributes to the ongoing dialogue on enhancing interpretability in AI systems. The quest to align the theoretical foundations of xAI with its practical applications accentuate a broader ambition: to develop interpretable and trustworthy AI systems that are accessible, understandable, and useable across diverse domains.

In conclusion, while this study marks a significant step towards understanding and evaluating xAI techniques, it also highlights the vast expanse of uncharted territory that remains. The call for standardization, alongside the development of adaptable and universally applicable xAI frameworks, emerges as the next milestones to hit in the quest for truly explainable AI. As the field advances, researchers and practitioners should work to foster collaboration, innovation, clarity, but also simplicity, ensuring that the future of AI is not only powerful but also comprehensible to those it serves.

CRediT authorship contribution statement

Marek Pawlicki: Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Formal analysis, Conceptualization. **Aleksandra Pawlicka:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Federica Uccello:** Writing – original draft, Software, Methodology, Investigation. **Sebastian Szelest:** Writing – original draft, Visualization, Software, Formal analysis, Conceptualization. **Salvatore D'Antonio:** Writing – review & editing, Supervision. **Rafał Kozik:** Writing – original draft, Visualization, Methodology, Formal analysis. **Michał Choraś:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

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Data availability

Data will be made available on request.

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